

F O R T I M A N A G E R N S E 6

Lab 08 — Importing Policy, ADOM Revisions and Conflict Recovery

01. The Brownfield Problem

Every lab so far has built policy in FortiManager and pushed it down. Real deployments rarely start that way. A site has firewalls running production policy for years, FortiManager arrives afterwards, and that existing configuration has to come up into the manager before anything can be standardised across the estate.

Import is the mechanism for that, and this lab builds the situation deliberately: a policy is created directly on the FortiGate, then brought into FortiManager.

The pair is unchanged from Lab 07 — FortiManager 7.4.3 and FortiGate 7.2.1 (fortigate7-2, 10.58.228.177) in the root ADOM.

► 1.1 A policy created on the firewall

The policy below was built in the FortiGate GUI, not in FortiManager. It reuses the LAN-Net address and the OPC-UA service from Lab 07, which are already present on the device because they were installed there.

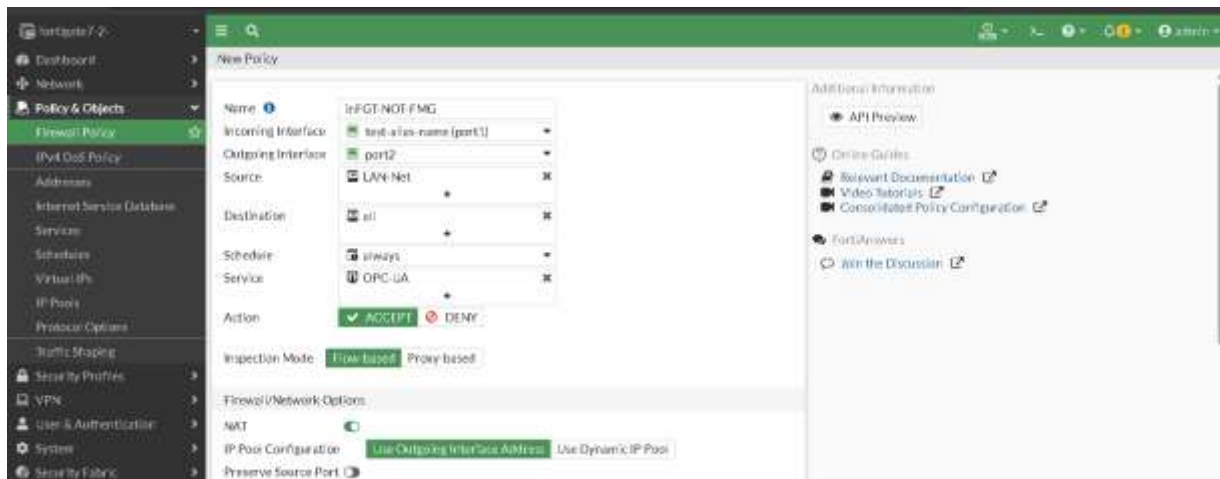


Figure 1 — inFGT-NOT-FMG created directly on the FortiGate — port1 to port2, action ACCEPT, NAT enabled.

Observation	This policy has Action ACCEPT and NAT enabled. The OPC-UA-POLICY built in FortiManager in Lab 07 has Action DENY.
Why	FortiOS defaults a new policy to ACCEPT in the GUI; FortiManager defaults to DENY. The same administrator building the same policy in the two interfaces gets opposite behaviour.
Check	This is worth remembering when comparing a device-built policy against a manager-built one. The difference is a default, not a decision.

► 1.2 Why it cannot simply be left there

Lab 05 established that FortiManager retrieves device-level settings automatically but does not retrieve policy, VPN or firewall objects. A policy created locally is therefore invisible to FortiManager and survives only until the next policy package install.

The policy is already scheduled for deletion

Nothing has gone wrong yet, and the policy is already doomed. The next time the package is installed, FortiManager will send the package as the complete policy set and the device-only policy will be removed.

This is what makes import a recovery step rather than a convenience. Section 03 shows FortiManager stating the deletion explicitly before it happens.

02. Importing the Policy Package

Import is run from the device record rather than from the policy package. FortiManager reads the device's live policy set and the objects it references, and writes both into the ADOM.

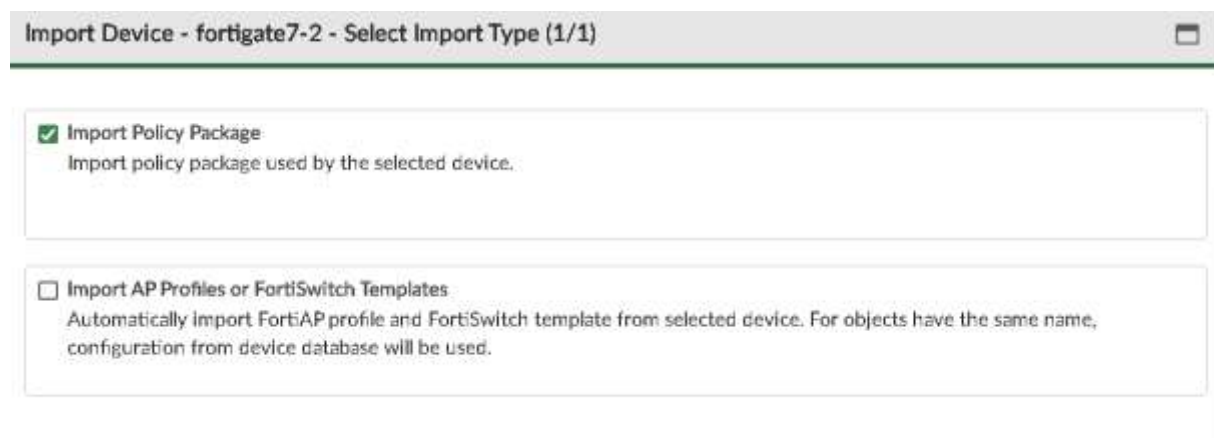


Figure 2 — Import Device — Import Policy Package selected.

The second option covers FortiAP profiles and FortiSwitch templates, and carries a note worth reading: where an object of the same name already exists, the device database version is used.

► 2.1 The import summary



Figure 3 — Policy Import Summary — 15 of 15 policies and objects imported.

Observation	The summary reports Firewall Policy 1 of 1, although the device was running two policies at this point — OPC-UA-POLICY from Lab 07 and the locally built inFGT-NOT-FMG.
Reading	Consistent with import bringing in only what the ADOM does not already hold. OPC-UA-POLICY originated in FortiManager, so only the device-built policy counted as an import.
Verify	The Download Import Report button on this screen produces the full per-object record. Worth keeping for a real migration, where the count is the only evidence of what did and did not come across.

▸ 2.2 Import creates its own package

The imported policy did not join the existing default package. FortiManager created a second package named after the device.

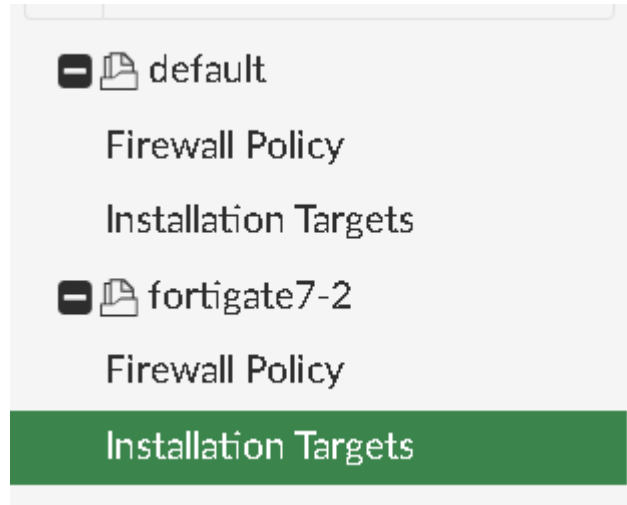


Figure 4 — Two policy packages — default, and a new one named fortigate7-2.

This matters more than it looks. The device now has two packages that could be installed to it, only one of which contains the imported policy, and the installation target list decides which one actually reaches it.

03. The Policy Package Diff

The Install Wizard offers a Policy Package Diff alongside the install preview. Where the preview from Lab 05 renders device settings as CLI, this compares the policy set in the package against the policy set currently running on the device, row by row.



Figure 5 — Policy Package Diff — one policy changed, one marked for deletion.

Row	Policy	What the diff says
Changed	OPC-UA-POLICY	Present in both, with differences — the destination reads LAN-Net in the package against what the device is running.
Deleted	inFGT-NOT-FMG	Present on the device, absent from the package. Installing this package removes it.

04. Deleting a Policy and Reviewing the Change

Removing a policy from a package follows the same staged pattern as every other change in FortiManager: the package is edited, the result is previewed, and nothing reaches the device until an install.



Figure 6 — Confirm Deletion for the selected policy.

The install preview renders the deletion as the CLI that will run on the firewall.

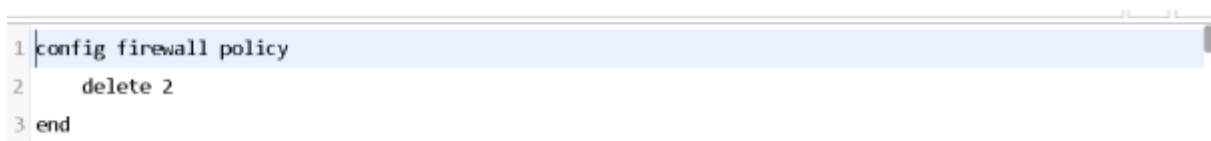


Figure 7 — Install preview — the deletion expressed as config firewall policy / delete 2.

Observation	The confirmation dialog names OPC-UA-POLICY, while the preview removes policy ID 2 — which is inFGT-NOT-FMG in the package diff in section 03.
Why	Policy IDs in a FortiManager package and policy IDs on the device are separate numbering. A package renumbers as policies are added and removed, and an import assigns new IDs.
Action	Read the preview by what it deletes, not by the name in the dialog. On a production package, confirming the ID against the device policy list before installing is the only reliable check.

05. ADOM Revisions

Lab 05 covered device revisions — a snapshot of one device's configuration. ADOM revisions are the other kind: a snapshot of the entire ADOM, including every policy package, every object and every per-device mapping in it.

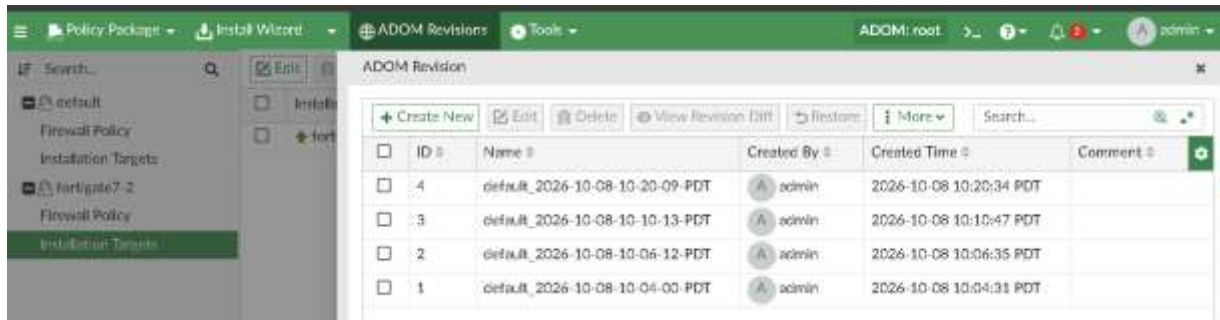


Figure 8 — ADOM Revision list with the revisions created during this lab.

Control	What it does
Create New	Takes a manual snapshot of the ADOM at this moment, with an optional comment.
View Revision Diff	Compares two ADOM revisions, or a revision against the current state.
Restore	Rolls the entire ADOM back to a revision — every package and object together, not one policy.

Restore is ADOM-wide

Restoring a revision rolls back everything in the ADOM, including work other administrators have done in packages unrelated to the problem being fixed.

It is the right tool after a bad bulk change and the wrong tool for undoing one policy. For a single policy, edit the package and install; the ADOM revision exists for the case where that is no longer practical.

Taking a manual revision before any significant change is the habit worth forming. The automatic revisions are created by the system on its own schedule, and the one that matters is rarely the one that happens to exist.

06. Conflict and Recovery by Retrieve

After a restart of the appliances, the device moved to Conflict — the same state produced by the failed install in Lab 07, reached here without any install having run.

Device Name	Config Status	Host Name	IP Address	Platform	Description	HA Status	Firewall
fortigate7-2	Conflict	fortigate7-2	10.50.228.177	FortiGate-VM64-L			FortiGate 7.2.1 Install View

Figure 9 — Device configuration status showing Conflict.

The diff between the device and the device database explains it.

```

edit "root_CA3"
    set ca "-----BEGIN CERTIFICATE-----
MIIDUzCCAjugAwIBAgIgM0MwQTRCNTk5REI4RTBCRTNENUM3Q0RDODkxMDQ5NTkw
DQYJKoZIhvcNAQEFBQAwKzEWMBQGA1UEChMNRM9ydGluZXQgTHRkLjERMA8GA1UE
AxMIRM9ydGluZXQwHhcNMjYwOTMwMTgwNzU3WWhcNMzYxMDA0MTgwNzU3WjArMRYw
FAYDVQQKEw1Gb3J0aW5ldCBMdGQUMREwDwYDVQQDEwhGb3J0aW5ldDCCASIwDQYJ
KoZIhvcNAQEBBQADggEPADCCAQoCggEBALwOi2Bkv1iSomio1+1xhbLEPnbTLJAN
ixvV865gX8ujzDOSAfkgPbWH3lqQt5VRj7bjKXOb74Ed52ZsvhnHgjaqfeKrHr1
BkBOqjZZrfbcTFOsK7nF7XZ2QNwzqxs3w3bzm2Ccf6GBDTs5+ewA06qlpNkFnch
CEMUpTQwjAPgISQj/4aeaxwzZcW0uKuu4EKspiVE+2WmgExX3upOtn30LUebJA
Owt2x1YKwU43Eyf8emfCXAYkdns2cEhIK2SL0GgFHNc6kcBwRReu/DWpb39DCBqV
SGl182yET13lbrwt2w+C/KMrjsW3uDzKoBhkyYw1OQLQyQg+T8BhZEsCAwEAAANj
MGEwHQYDVR0BBYEFJlLrd8VCR04/5hrh4jnfXMX9+/uMB8GA1UdIwQYMBaAFJlL
rd8VCR04/5hrh4jnfXMX9+/uMA8GA1UdEwEB/wQFMAMBAf8wDgYDVR0PAQH/BAQD
AgGGMAGCSqGSIb3DQEBAUAA4IBAQBtQ6vwIeXOIvMh3Shxt9hvO6YhkLSOLbC3
ogAcr7YYP2i0JD6sSdzjn/uXkdFGQymrm8Y1/R7f2AroIbAMXrMWjDEdvA2oCUsR
UOFpd2GZzg1N62GmDCM4jr5ShbPh2Rfko1Inr/H4KI8O/r0DdsiAgpf5KW0teDRn
SmJQGjs1lL1fi0cnuXcK45Hy1aWRuXWbba2dGnPHoxvlf0narWQGxqIsejXdqrXn
mnposFdDS6hIFMPsP3byj+CnMW8cjBAK2lN0/TefT429yNskEiJefS60WwPUq0v4
qQkSdIjEVB1rLAON9cob2+z+Hbb/W/SAdpYi2pkvtzbP6ZSPZxu1
-----END CERTIFICATE-----"
    set range global

```

Figure 10 — The differing content — a CA certificate block under config vpn certificate ca.

Observation	The difference is a certificate authority entry, root_CA3, with its full PEM body and a global scope — not policy, not objects, not anything configured in this lab.
Reading	A reboot on its own does not rewrite the certificate store. What does is the device refreshing its bundled CA list at startup, which adds or updates entries the device database captured before the refresh.
Effect	The two copies disagree on content no administrator touched, and FortiManager reports that as Conflict because it cannot tell an intentional change from an automatic one.

► 6.1 Configuration Revision History

Retrieve is reached from the device's revision history, which is also the clearest record of everything the last two labs have done to this device.

ID	Date & Time	Name	Created by	Installation	Comments
18	2026-10-08 10:21:17	default	admin	Failed	
17	2026-10-08 10:17:10	fortigate7-2	admin	Failed	
16	2026-10-08 10:07:17	default	admin	Failed	
15	2026-10-08 10:05:09	default	admin	Failed	
✓ 14	2026-10-08 09:51:59	fortigate7-2	admin	Installed	
13	2026-10-07 12:06:16	fortigate7-2	admin	Installed	
12	2026-10-07 11:53:47	fortigate7-2	admin	Installed	
11	2026-10-07 11:46:10		script_manager	Retrieved	
10	2026-10-07 11:44:50	fortigate7-2	admin	Installed	
9	2026-10-07 11:41:06		script_manager	Retrieved	
8	2026-10-07 11:35:43	fortigate7-2	admin	Installed	
7	2026-10-07 11:31:34	AutoUpdate	AutoUpdate	Auto Updated	Auto update from device
6	2026-10-07 11:29:24	fortigate7-2	admin	Installed	
4	2026-10-07 11:24:34	fortigate7-2	admin	Installed	
2	2026-10-07 11:18:53	fortigate7-2	admin	Installed	
1	2026-10-07 11:11:22	fortigate7-2	admin	Installed	

Figure 11 — Configuration Revision History — eighteen revisions with their outcomes, and the Retrieve Config control.

Installation value	What created the revision
Installed	A successful install from FortiManager. The green tick marks the revision currently in effect.
Failed	An install that did not complete. Revisions 15 to 18 are the failed attempts from Lab 07 and this lab — a failed install still writes a revision.
Retrieved	A configuration pulled up from the device. Created by script_manager here, as a side effect of script execution in Lab 06.
Auto Updated	The automatic retrieval from Lab 05, carrying the comment "Auto update from device".

► 6.2 Retrieving

Retrieve pulls the device's current configuration into the device database, overwriting what the database held.

Retrieve Device Revision

100%

Total: 1/1, Success: 1, Warning: 0, Error: 0 Show Details

[View Progress Report](#)

#	Name	Time Used	Status
1	fortigate7-2	1m 20s	Retrieve finish

Figure 12 — Retrieve Device Revision — completed in 1 minute 20 seconds.

Because nothing in the database was wanted here — the only difference was a CA bundle that originated on the device — the retrieve resolves the conflict cleanly.

Device Name	Config Status	Host Name	IP Address	Platform	Descriptors	Firmware Version
fortigate7-2	Synchronized	fortigate7-2	10.58.228.177	FortiGate-VM64...		FortiGate 7.2.1 build1254 (GA)

Figure 13 — Configuration status back to Synchronized.

Retrieve discards the database side

Retrieve is safe here because the device holds the correct content. It is not a general conflict fix: any pending change staged in the device database is overwritten by it and lost.

Before retrieving on a device in Conflict, check whether the database holds work that has not been installed. If it does, the difference has to be reconciled first — retrieve is the resolution for a conflict the device caused, not for one the administrator caused.

07. Detecting Policies After the Conflict

Retrieve resolved the device status, and it also changed what FortiManager knows. The device database now holds the device's full policy set, while the default package holds its own — and the gap between them is wider than it was before.

7.1 The diff after the retrieve



Policy Package Diff (fortigate7-2)

Diff Preparation Total: 2/2, Success: 2, Warning: 0, Error: 0

Summary firewall policy

firewall policy - deleted (2)

	#	Policy ID	Name	From	To	Source	Destination	Schedule
Deleted	1	1	OPC-UA-POLICY	"any"	"any"	"LAN-Net"	"LAN-Net"	"always"
Deleted	2	2	InFGT-NOT-FMG	"port1"	"port2"	"LAN-Net"	"all"	"always"

Figure 14 — Policy Package Diff — both policies now marked for deletion.

Observation	The diff in section 03 showed one policy changed and one deleted. It now shows both OPC-UA-POLICY and inFGT-NOT-FMG as Deleted.
Why	The retrieve rebuilt the device database from the firewall, so the database and the device now agree on two policies. The default package contains neither of them in the form the device is running, so installing it would remove both.
Reading	Synchronized describes the device database against the device. It says nothing about whether a policy package matches either of them — those are separate comparisons, and this diff is the second one.

► 7.2 Re-importing into a new package

Rather than reconcile the default package by hand, the device's policy set was imported again into a package of its own.

Import Device - fortigate7-2 - Interface Mapping & Policy (2/5)

Create a new policy package for import.

Policy Package Name:

Folder:

Policy Selection:

Object Selection:

ⓘ When importing configuration from this device, all enabled interfaces require a mapping to an ADOM Level interface. Note, the same ADOM Level interface can map to different interfaces on the each device.

Device Interface	Mapping Type	Normalized Interface
port1	Per-Device <input checked="" type="button" value="Per-Platform"/>	port1
port2	Per-Device <input checked="" type="button" value="Per-Platform"/>	port2

Add mappings for all unused device interfaces ☒

Figure 15 — Import Device step 2 — package name, policy selection, object selection and interface mapping.

This step carries three decisions the first import in section 02 did not surface.

Setting	Choice made	What it means
Policy Selection	Import All (3)	Every policy on the device comes across. Select Policies to Import allows a subset — useful when migrating a firewall whose policy set is partly obsolete.
Object Selection	Import only policy dependent objects	Only objects referenced by the imported policies come across. Import all objects brings the device's entire object table, including anything unused.
Interface mapping	Per-Platform, port1 to port1 and port2 to port2	Device interfaces are mapped to ADOM-level normalized interfaces so a policy can reference one name across devices whose physical ports differ.

Normalized interfaces are the interface equivalent of per-device mapping

A policy references a normalized interface; each device maps that name to its own physical port. Per-Platform maps by hardware model, so every FortiGate of the same platform inherits the mapping; Per-Device maps one named device.

This is what allows one policy package to install to a branch firewall whose WAN is port1 and a site firewall whose WAN is wan2, without editing the policy.

7.3 The imported package

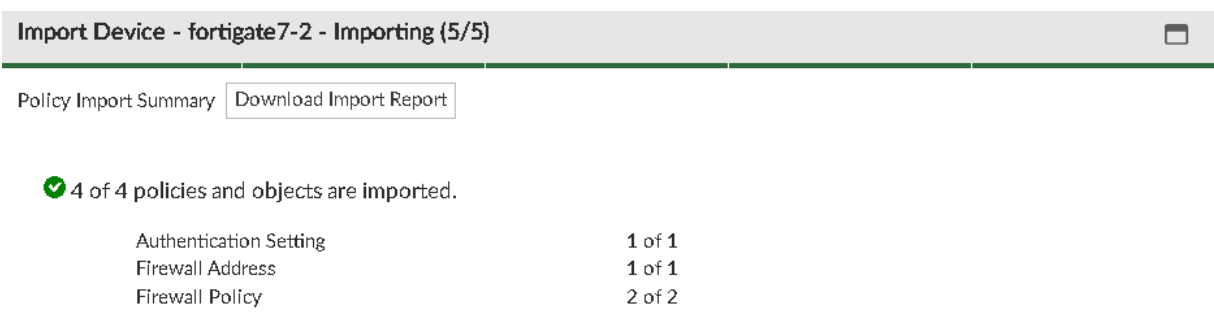


Figure 17 — Policy Import Summary — 4 of 4 policies and objects imported, including both firewall policies.

Both policies are now in the package, with the actions they were built with — OPC-UA-POLICY denying and inFGT-NOT-FMG accepting — above the implicit deny.

#	Name	From	To	Source	Destination	Schedule	Service	Action	Security
1	OPFC-UA-POLICY	any	any	LAN-Net	LAN-Net	always	OPFC-UA	Accept	hnd
2	INFOS-INT-FMG	port1	port2	LAN-Net	all	always	OPFC-UA	Accept	hnd
3	Implicit Deny (UI/3 Times)	any	any	all	all	always	All	Deny	hnd

Figure 18 — The new package fortigate7-2-after-Conflict holding both policies.

Three packages for one device

The ADOM now holds default, fortigate7-2 from the first import, and fortigate7-2-after-Conflict from this one. Only the package named in the device's installation targets will reach the firewall.

Each import creates a package rather than updating one, so repeated imports during a migration leave a trail of near-identical packages. Deleting the superseded ones is part of finishing the migration, not optional tidying.